Does Okun's Law hold in the AI Era?

Is the historical correlation between economic output and unemployment weakening because of AI?

For sixty years, economists have relied on a rule known as Okun's Law that asserts that when GDP rises above long-run potential, unemployment will fall, and when growth slows, unemployment rises.

This research asks whether the rise of generative AI (using ChatGPT's late-2022 release as a useful natural marker[1]) has allowed firms to produce more output without proprietary hiring of workers. If yes, then a central tool of macroeconomic policy is becoming unreliable, and the application of monetary and labor policy, as well as education, is significant.

In 192, Arthur Okun noticed that every 1 percentage point growth of real GDP above its sustainable trend was associated with roughly a .5 percentage point fall in the unemployment rate. The exact magnitude has been estimated many times, but its negative relationship has remained a core pillar of economics.

Of course, it seems intuitive that when demand for goods and services rises, firms need more labor to produce them, so they hire, and unemployment falls. However, has it recently become possible that labor is optional for additional optional causing a pivotal break in the once status quo of economics.

Data ey

Series	Content	Souce	Use
GDPC1	Real GDP (2017 dollars)	BEA, quarterly	Econoimc output
GDPPOT	Potential GDP- what GDP would be at max sustainable	CBOW quarterly estimate	Reference for normal output

[1] However this effect may not have actually started to have a noticeable impact until more than a year later when its capabilities became widely recognized and employed within modern society

	output		
UNRATE	Civilian unemployment rate	BLS, monthly (resampled to quarterly mean)	Actual unemployment
NROU	Natural rate of unemployment	CBO estimates quarterly	Unemployment reference line

Charts:

It doesn't make sense to compare raw GDP across decades because a 1 billion dollar deviation now versus the 90s would mean completely different things. To compare across eras, both GDP and unemployment are converted into deviations from normal.

Y(gap)=(GDPC1-GDPPOT)/GDPOT *100

Positive→ economy producing above potential output
Negative→ producing below

Unemployment Gap

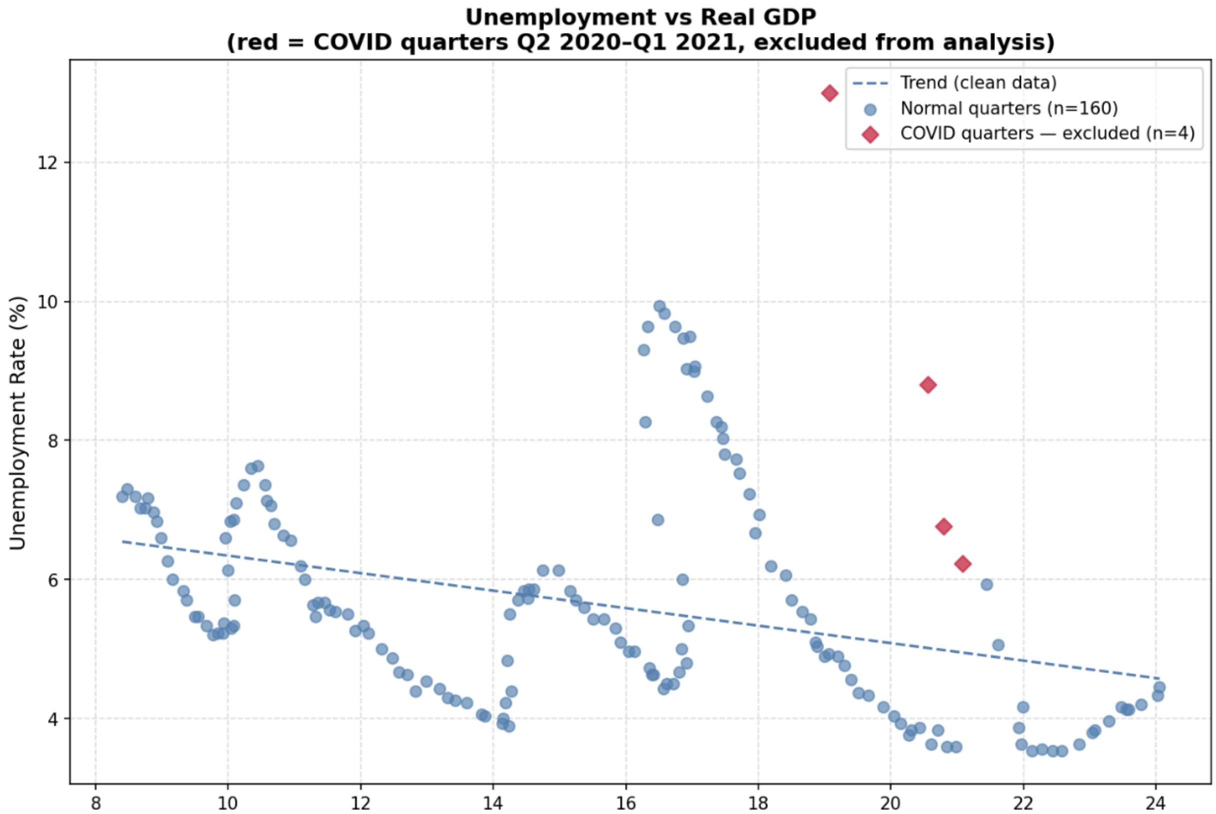
Ugap = UNRATE-NROU

Positive → more unemployment than NROU
Negative→ less unemployment than NROU

Excluding covid

We decided to exclude all data from Q2 of 2020 to Q1 of 2021, as during these quarters, many businesses were legally closed, and people were legally prohibited from working. The spike in unemployment and the collapse in GDP were created in a way that is unnatural to a normal economic cycle, and they reflected a global pandemic that, in all likelihood, will not be seen again for the extent of our lifetimes.

The COVID quarters are kept in the dataset for plotting (shown as red diamonds in Chart 1) so exclusion is transparent, but they are removed before any slope, correlation, or rolling coefficient is computed.

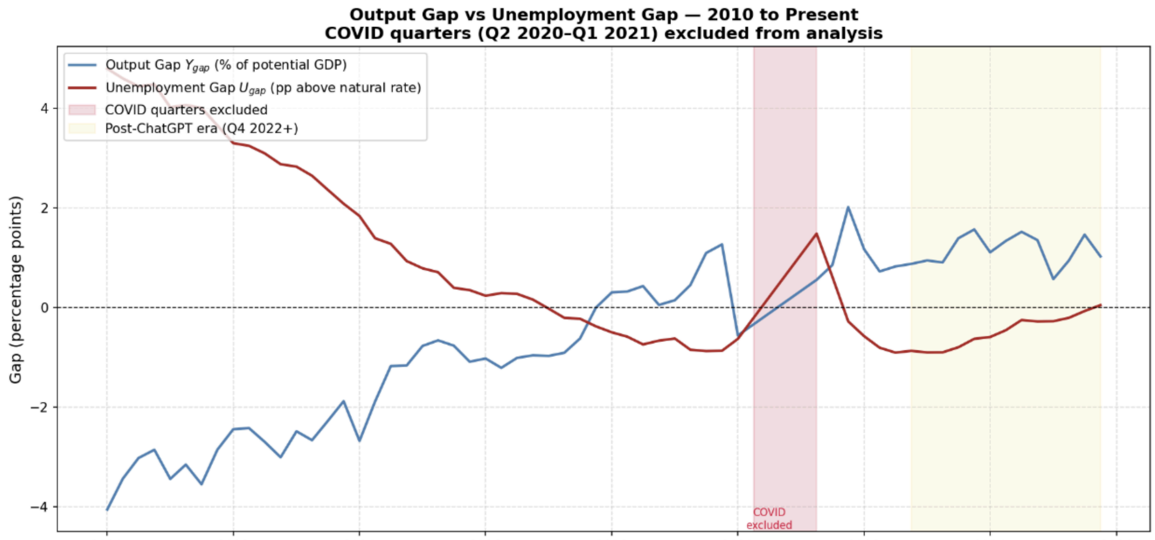


A scatter where every dot is one quarter of U.S. history. The X axis is real GDP (in trillions of 2017 dollars); the Y axis is the unemployment rate.

Left to right means reading forward in time.

This chart illustrates the relationship between rGDP and unemployment, as well as why COVID has to be removed before any serious analysis.

Limitations:
Mixes 1980s, 200s, and 2020 data with no time discrimination

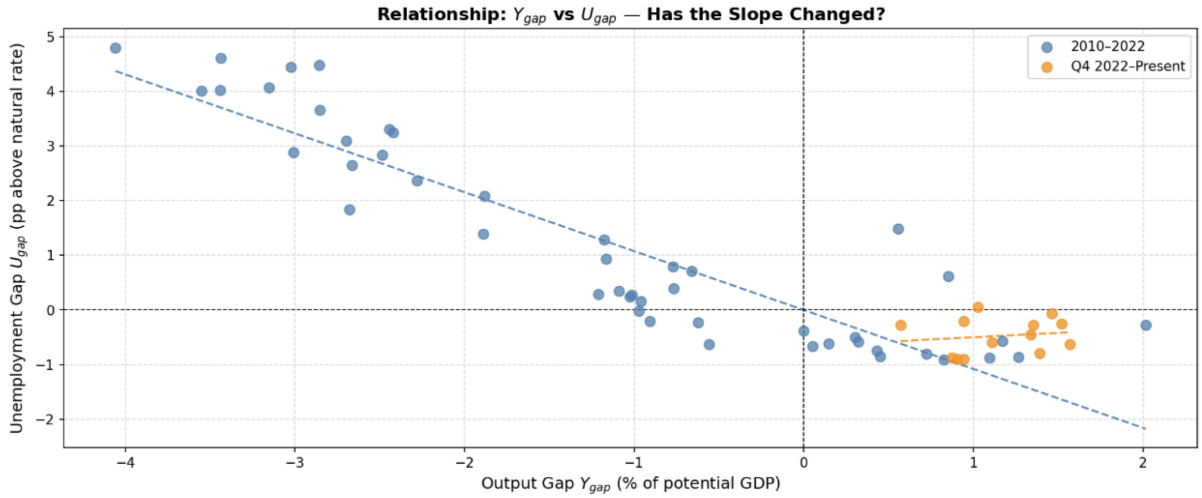


Our (ygap) output gap is in blue, and the unemployment gap (Ugap) is plotted in red on the same axis from 2010. The pink stripe marks excluded COVID quarters. The yellow stripe marks the post-ChatGPT era (Q4 2022 onward).

It is expected that when Ygap rises, the Ugap should fall (the two lines should essentially mirror each other across the zero line).

From 2010 to 2019, the lines mirror with a very high inverse correlation. As the RGDP recovered from the Great Recession (rising blue line), the unemployment gap predictably diminished (red line falling). Here, as Okun predicts, every percentage point of output recovery is converted into job creation.

Beginning in late 2022, something visually new appears: the output gap stays clearly positive (1. To 1.5%), but the unemployment gap fails to move significantly negative. The two lines flatten and run nearly parallel rather than mirroring one another.



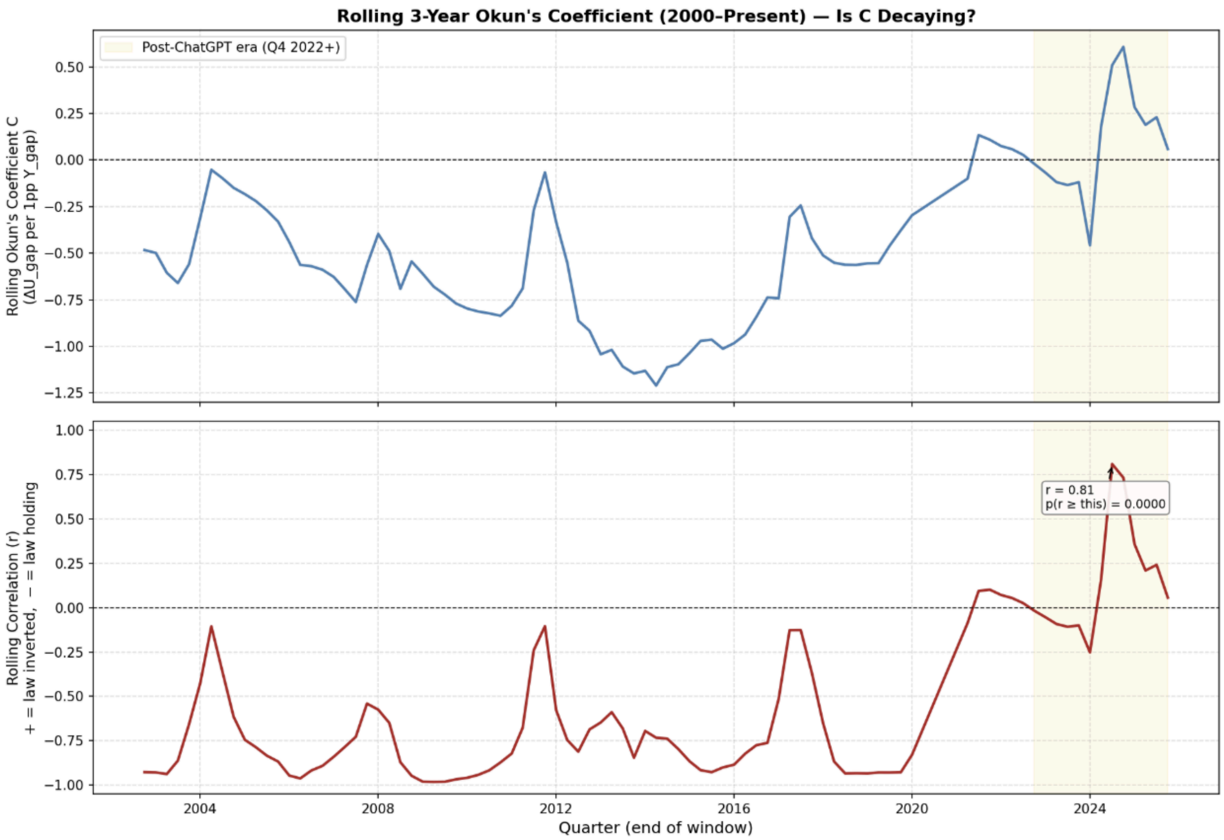
Each dot is one quarter from 2010 onward, plotted as (Ygap, Ugap). Blue dots are 2010- Q3 2022; orange dots are Q4 2022 present. A best-fit line is drawn through each cluster.

The blue cluster shows a clearly and consistently negative slope as Olun's law would predict. The orange cluster sits in the bottom right quadrant (it shows data points from only Q4 2022 to present) and shows essentially no slope.

Limitations

The orange cluster is only 12 quarters; the slope estimates from small samples are noisy.

The post CHatGPT cutoff is useful but somewhat arbitrary date; ChatGPT was released on Nov 30, 2022, but enterprise AI adoption happened more gradually.
Doesn't yet establish causation; it only documents the reek.s



2022):

- Over 76 rolling windows before the AI era, r averaged **-0.689** with a std of **0.307**
- Only **6.6%** of those historical windows ever had a positive r — and even those were barely above zero during recessions

The post-ChatGPT windows:

This chart runs a sliding 12-quarter (3-year) regression that estimates the Okun coefficient C at every quarter.

Top panel

For each quarter on the x-axis, the line plots the slope of the regression Ugap = C *Ygap + intercept estimated on the trailing 12 quarters.

C< .5 Okun's Law holds (change in unemployment reflects an even greater change in the output)
0<C<.5 The relationship is weaker, but still holds

C>0 The relationship has inverted

From 2000-2019, C hovers between 0 and -1. The sign is consistently negative. Okun's law is doing exactly what it's supposed to for nearly two decades straight.

After Q4 of 2022 (Yellow shading): C swings erratically, briefly reaching above + .5 than collapsing back towards zero, highlighting a significant loss in stability

The Bottom Panel shows a Rolling R correlation.

It shows the Pearson correlation between Ygap and Ugap.

For most of 200s and 2010s, r sits near -1; the two gaps were almost perfectly negatively correlated in the post-ChatGPT window, r rises sharply and peaks at +.81.

Tested against the historical distribution of pre 2022 r values (excluding COVID treated as roughly normal with the sample's mean and standard deviation), the probability of observing an r this positive by chance is:

p(r>.81)= 0.0000

Why this matters

1. The sign of Okun's relationship has flipped in some recent windows
2. The magnitude has become unstable
3. The inversion cannot just be seen as random; it is far enough into the tail of the historical distribution that conventional statistics reject the null hypothesis.

Key limitations of this chart:

- A 12-quarter window is only 3 years of data; short windows can be very "noisy."
- The pre- 2022 windows are correlated with each other (overlapping), which technically violates the i.i.d assumption behind the p-value; the true P is somewhat larger than 0.0000, though in all likelihood nearly as tiny.
- Some of the post - 2022 instability could reflect post-pandemic labor market peculiarities rather than AI (this will soon be addressed)

(Chat suggestion: **Ways to extend this chart.**

- Try multiple window sizes (8, 12, 16, 20 quarters) and show all in a small-multiples grid — robustness check.

- Use bootstrap resampling instead of a normal-distribution approximation for the p-value.
- Add a confidence band around CC C using each window's regression standard error.
- Estimate the same coefficient using **growth-rate Okun's Law** (changes in GDP vs changes in unemployment) as an alternative specification.

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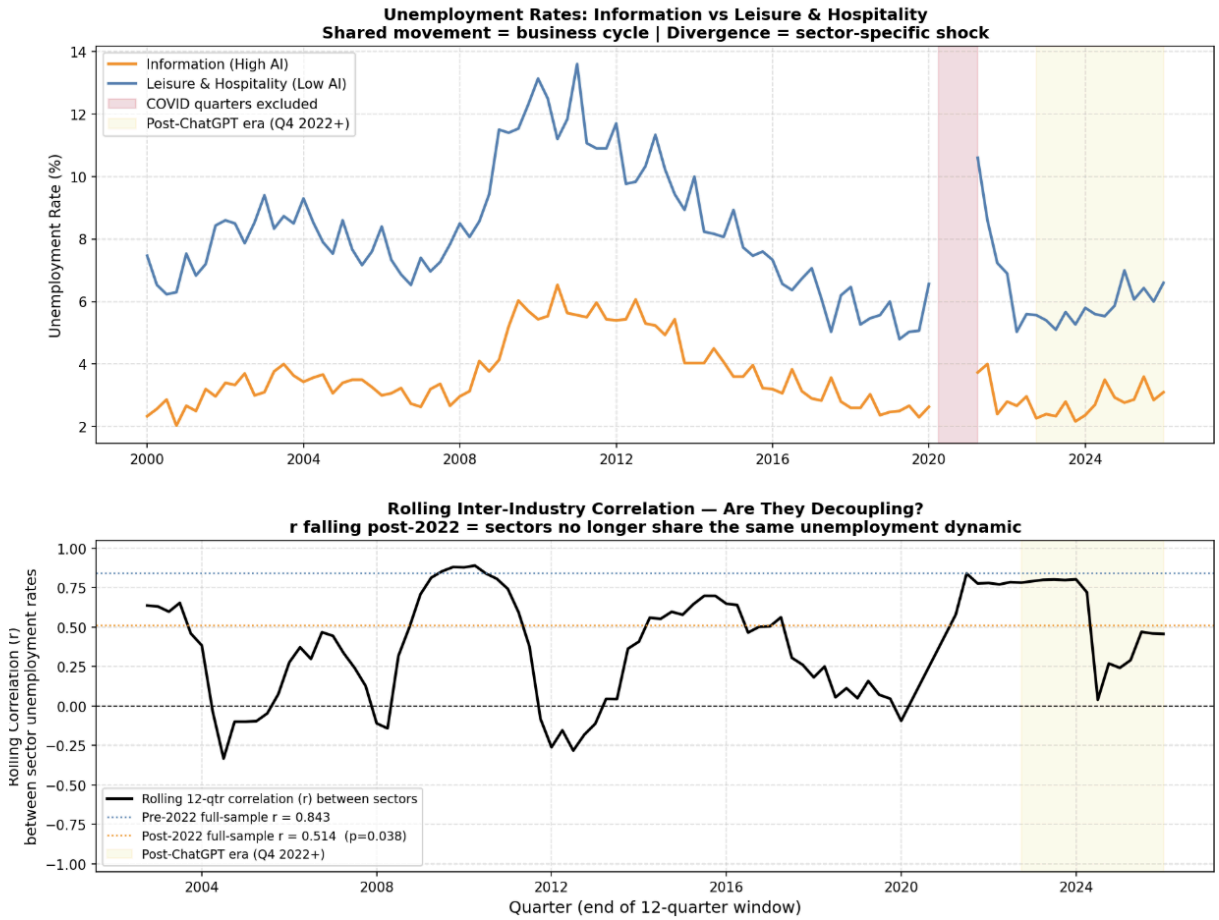
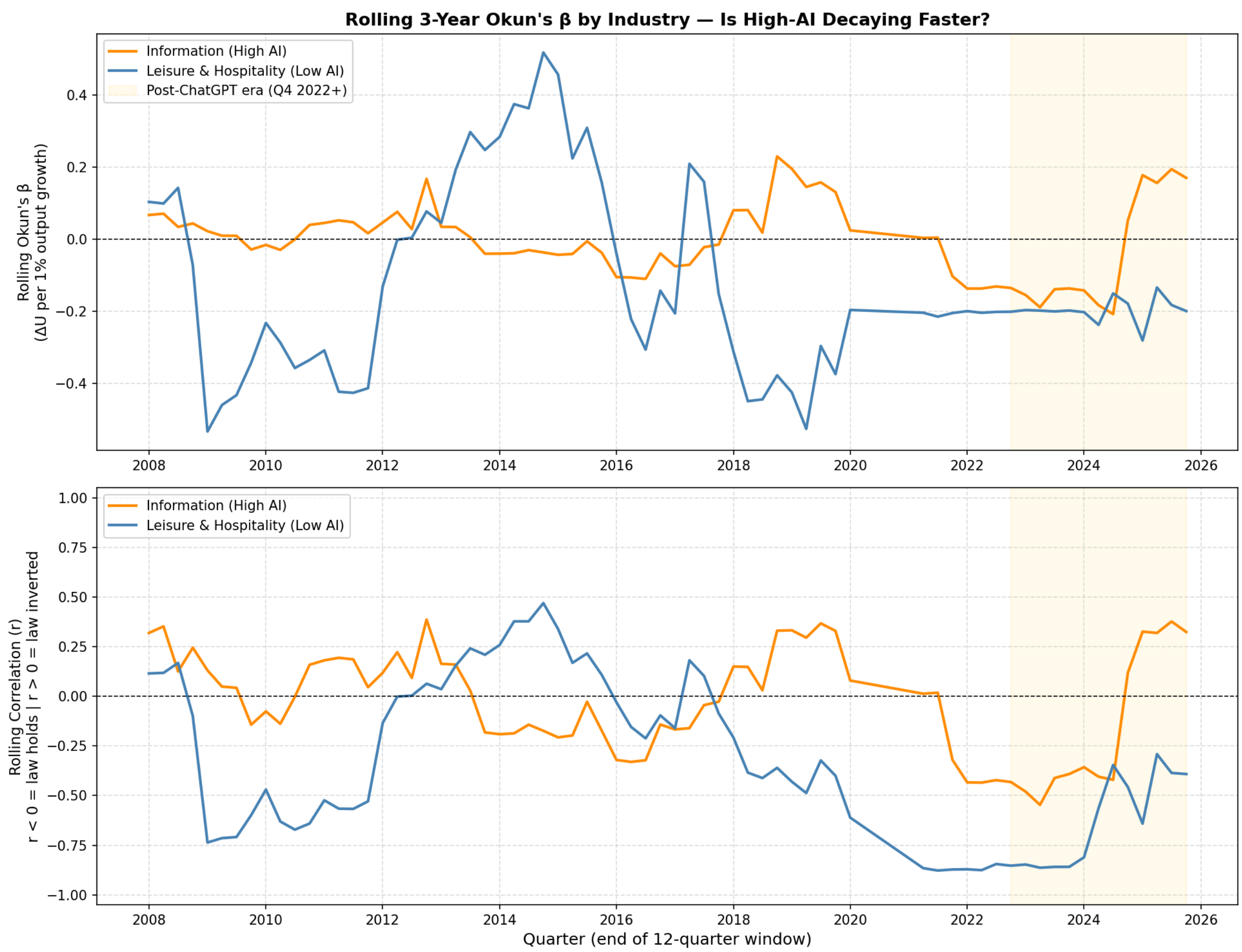


The top panel shows unemployment rates by Industry. If the aggregate Okun's breakdown is actually drive by AI it should show up most strongly in industries most exposed to AI, and weakly or not at all in industries that are less automatable.

The **top panel** shows quarterly unemployment rates from year 2000 onwards for both sectors

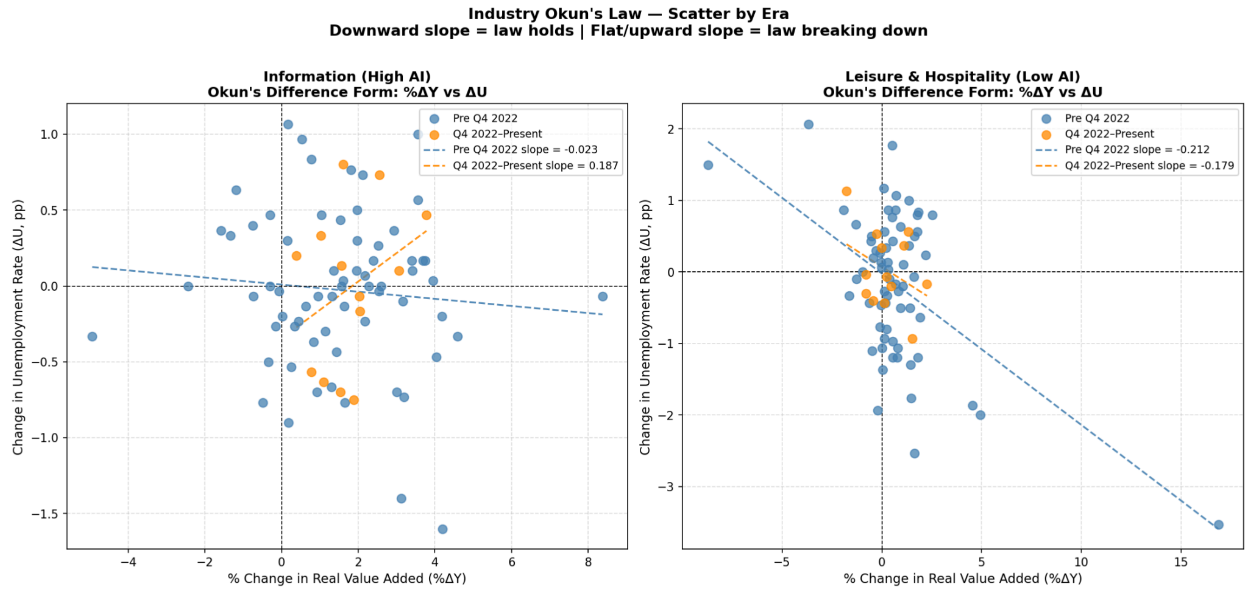
The bottom Panel estimate a 12 quarter correlation between the two industries' unemployment.

However, what this chart doesn't show is that the output of the information sector significantly increased while that of the hospitality sector barely changed.
(**<u>Strangely are not decoupling as much as our hypothesis would state low probability, but may be in large part due to such a short period for the 12-year rolling correlation)</u>**



The top panels show a 3-year Okun's rolling coefficient by industry (how much unemployment will change for a one percent change in output, so inverted), while the Leisure & Hospitality industry remains as what Okun would predict.

The rolling reinforces this pattern from a probabilistic perspective, where a negative r shows that Okun's law is holding as expected, and a positive r shows that it is inverted.



This graph shows each quarter as a point and highlights the differences in slope of the two industries side by side. Post 2022 in the Information industries, each quarter is extremely variable in the predicted Okun's law, forming an even slightly positive slope. However, in the Leisure & Hospitality Industry, the sope (relationship between GDP and unemployment remains predictably negative throughout.

The key limitations of this chart are that industry-level data is far more volatile than aggregate GDP.
Two industries are a small comparison set; in the future, we will go industry by industry to better view this relationship, and other low-AI sectors might tell different stories.

Since Q4 of 2022, the U.S> has experienced a sustained period of above-potential output without a proportional decline in the unemployment gap. The Okun coefficient has become unstable for the first time since 2000, and the rolling Y-U correlation has become inverted in a way that has near-zero historical probability. The breakdown is concentrated in AI-exposed industries and largely absent in non-AI-exposed industries, which is the pattern AI-driven productivity gains would predict. The evidence is consistent with, but does not yet prove, a structural shift in the output-employment relationship driven by AI.

The nine-industry test was built to answer a specific question that the two-industry comparison couldn't: if Information's Okun coefficient flipping positive after late 2022 was really about AI rather than a coincidence, then industries with more theoretical exposure to AI should show bigger flips, and industries with less exposure should show smaller ones or none at all. Felten, Raj, and Seamans' AI Industry Exposure score gave a ready-made way to rank nine industries on exactly that dimension, so the natural next step was to plot each industry's change in its Okun coefficient against its exposure score and see whether a line fit through the nine points sloped the way the hypothesis predicted.

Before that comparison could be trusted, though, a labeling error in the chart needed catching. The scatter plot's axis description claimed a more negative change in the coefficient meant the law had weakened more, but Construction's own numbers show the opposite: its coefficient moved from a strongly negative −0.386 before 2022 to a barely positive +0.046 after, which is unambiguously a weakening, and that's a positive change, not a negative one. Once that sign convention was corrected, the regression underneath the scatter — a real correlation of −0.68 with a p-value of 0.045 across the nine industries — turned out to mean something uncomfortable: higher AI exposure predicted the Okun relationship getting *stronger*, not weaker. That's the reverse of the hypothesis, and it's a strong enough result to take seriously rather than wave away.

Looking at the nine industries individually, only two — Information and Wholesale Trade — actually behaved the way the AI story predicts, with high or moderate exposure scores and coefficients that flipped toward positive after 2022. Three industries with genuinely high exposure scores — Financial Activities, Professional & Business Services, and Education & Health — showed the opposite: their Okun relationships held steady or got stronger even though their theoretical AI exposure was among the highest in the sample. The remaining four, mostly low-exposure industries like Construction, Manufacturing, and Transportation & Utilities, showed the largest breakdowns of anyone in the dataset, which is backwards from what a dose-response relationship should look like if AI exposure were the active ingredient.

Three of those industries added useful texture once examined individually. Construction's correlation flipped from a tight −0.725 before 2022 to an equally tight +0.775 after, but its actual coefficient barely moved off zero in either direction — meaning the relationship between its output and unemployment became very reliable and predictable post-2022 without becoming economically large, a distinction the correlation number alone obscures. Its pre-2022 fit was also visibly anchored by a handful of extreme data points from the 2008–2012 housing crash, which raises a real question about how much of that −0.386 baseline reflects one historical crisis rather than a stable long-run pattern. Education & Health showed a weak relationship in both periods, never exceeding a correlation of about 0.2 in either direction — unsurprising for a sector whose employment is driven by demographics and funding cycles rather than the business cycle, and one whose underlying data has a known mismatch between the population covered by its employment series and the narrower population covered by its output series. Financial Activities turned out to be the most interesting case: its real output has nearly doubled relative to its 2019 level while its unemployment rate has stayed low and flat since roughly 2013, a genuine and dramatic divergence between output and jobs — but one that's been building gradually for over a decade rather than appearing suddenly in late 2022, which is exactly why a test built to detect a sharp break around a single cutoff date found nothing unusual there.

The biggest threat to a clean AI interpretation, though, is timing. The Federal Reserve's most aggressive rate-hiking cycle in roughly forty years began at almost the same moment as the AI cutoff used throughout this analysis, and the industries that showed the most dramatic breakdowns — Construction, Manufacturing, Transportation & Utilities — are also the most interest-rate-sensitive industries in the economy. A test that can't tell those two events apart will attribute rate-driven disruption to AI by default, simply because they happened at the same time.

Put together, this splits the original thesis into two separate claims with two separate fates. The claim that Okun's Law weakened in the aggregate U.S. economy since late 2022 is untouched by any of this — that evidence stands on its own. The claim that AI specifically caused that weakening is now genuinely uncertain rather than established, because the one test built to support it, using this particular measure of exposure, came back pointing the other way. That's not the same as the AI mechanism being false; it's evidence that this specific test, with this specific proxy and this specific confound sitting on the cutoff date, can't currently distinguish AI from monetary policy.